Housing First or Treatment First? Ask the Kids, Not the Parents

Manuscript RED-D-26-00078

Response to Referee 3

Major and minor concerns, responses in red

Thank you for the report, and there is no need to apologise for the timing. Your question at M2(c), where the case for intervention comes from if drug use is rational, is now the question the paper is organised around.

One note before the comments. The paper has a new title: it was submitted as *Housing First or Treatment First?* and is now *Housing First or Treatment First? Ask the Kids, Not the Parents*. The reason is the substance of the revision rather than a relabelling, and it will be clearest at M2(c), which asks where the externalities come from.

What changed that matters for your comments. The paper now asks whether housing assistance, in the programme as it exists and therefore rationed, should require sobriety, and it begins by asking whether moral hazard alone justifies the condition. It does not. With every effect of a parent's drug use on its child removed, a newborn is worse off under Treatment First, by 1.09 percent of lifetime consumption, even when the savings of the screen (Treatment First's drug test, which revokes the voucher of a user it catches) buy more vouchers for other households. The answer turns instead on the children, through three externalities that a parent counts only through an imperfect weight on its child. A parent's drug spending and lost earnings are consumption and housing its child goes without (the budget externality); a child whose parent had a substance use disorder (SUD) while it lived at home starts adult life with lower productivity (the earnings externality); and a child lives wherever its parent lives (the childhood homelessness externality). In the long run, with the drug price clearing its market and the voucher probability balancing the budget, Treatment First cuts drug use by 21 percent and homelessness by 11 percent, and a newborn gains 2.57 percent of lifetime consumption, three quarters of it an externality its parent does not count. The parents gain too, by 0.69 percent of their own lifetime consumption. The transition is now solved for every cohort alive at the reform, with the drug price and the budget clearing every period (Sections 6.4 and 6.5). Along it, homelessness and childhood homelessness never rise, and children born at the reform gain 1.45 percent.

Several of the changes answer your comments directly. Preferences are the same in every state, so reducing the disorder no longer raises the utility of consumption by construction (M2(a)). A new two period model (Section 3) shows that it is the uninsurable mortality risk of a SUD that puts drug use at the bottom of the income distribution, and the quantitative model is now set against the whole distribution of use by income in the data (M1(c), M2(b)). The drug price clears a supply curve, and the demand elasticities are checks rather than targets (M2(d)). The loss of consortium is gone (m2). Two things you asked about are no longer in the paper, and I say why where they come up: labour supply is now inelastic (m1), and the experiments no longer move from rationed to universal vouchers (m3).

Major concern 1: connection to empirical evidence and stylized facts

M1(a) Present stylized facts prominently

> *"The nature of the model, with its large number of parameters, makes it relatively opaque. I believe it would be helpful for the paper to include a few stylized facts more prominently, notably on the empirically observed correlations between drug use, voucher receipt, and homelessness."*

You are right that the facts were hard to see behind the model. I have not added a separate empirical section. Instead, each fact now sits where it is used, next to the model's counterpart, so the reader sees both at once.

- **Drug use and income.** Section 3 opens with the two facts the stylized model is built to explain. A user in the paper is an adult with a drug use disorder of any severity, mild included, and fewer than one in ten adults are users: 8.90 percent of adults aged 26 and over in the 2024 National Survey on Drug Use and Health (NSDUH). And the average user's income is 0.74 of the population average. Figure 2 now shows the whole relationship (M1(c)).
- **Drug use and homelessness.** A SUD, meaning a moderate or severe disorder, is far more common among the homeless than among all adults: 21.7 percent of the homeless have one (Gutwinski et al., 2021), against 3.68 percent of adults aged 26 and over in the 2024 NSDUH. The same holds for serious mental illness (SMI): 25 percent of the homeless, against 4.82 percent of adults in the 2024 NSDUH. The shares among the homeless are calibration targets (Table 3, Blocks 2 and 3). Table 4 adds the age profile of the homeless, how many of them work and what they earn (Block 1).
- **Voucher receipt.** 26.6 percent of income eligible households hold a voucher in the Picture of Subsidized Households of the Department of Housing and Urban Development (HUD) for 2023, the target for the probability of drawing one (Table 3, Block 2). Table 4, Block 3, adds what holders pay in rent and how many live mainly on wages.

[Still open.] One correlation you name is not in the paper: the paper does not report drug use among voucher holders, in the data or in the model.

The Associate Editor asked (A1) that the stylized model generate the facts it starts from. Section 3 does this for drug use and income, which decides whose children bear the costs of drug use.

The facts are in the opening of Section 3, in Figure 2 (Section 5.2, Block 2), and in Tables 3 and 4.

M1(b) Survival and unemployment probabilities are not transparent

> *"Survival probabilities (especially their dependence on homelessness and drug use) and unemployment probabilities are an essential input into the model's calibration and welfare implications. It would be good to present more prominently some empirical evidence on their correlation with homelessness and drug use, which are currently borrowed from other papers but not always set transparently (for example for unemployment transition probabilities)."*

Both are now set out in one place in the model, Section 4.8, which takes the death risk, the unemployment risk and the child's productivity penalty in turn. Every multiplier has its source in Table 3 or Table A1, with the model's value beside the data wherever the parameter is calibrated.

Survival. Baseline survival by age is the 2010 life table for the total population of the United States (Arias, 2014), for ages 26 to 85 (Appendix A, Block 10). Three multipliers raise the probability of dying, equation (20):

- a SUD, by $\kappa_{\phi d} = 4.00$. It is calibrated to the income of users relative to the average, because it is the parameter that decides how far down the income distribution use sits (Section 3.3). The value lies within the mortality rate ratios of a SUD in the literature: 3.91 in the Danish registers (Plana-Ripoll et al., 2019) and 6.6 among Norwegians in specialist care for a SUD (Heiberg et al., 2018).
- homelessness, by $\kappa_{\phi H} = 1.60$ at every age, the mortality ratio of homeless adults against poor housed adults in Meyer et al. (2023), so that the comparison is with households of similar means.
- SMI, by $\kappa_{\phi m} = 2.50$, the middle of the standardised mortality ratios for SMI in Olfson et al. (2015).

Two untargeted checks bear on them (Table 4). Excess deaths due to a SUD are 14.5 percent of deaths at ages 25 to 44 in the model, against 20.0 percent from drug poisonings in the WONDER database of the Centers for Disease Control and Prevention (Block 2). And losing a voucher raises the three year mortality of households with SMI by 3.6 percentage points in the model, against the 4.6 points by which enrolling in Housing First lowers it among homeless veterans with mental illness in Costantini (2025) (Block 7).

Unemployment. Equation (21) sets out the transitions. The baseline separation rate is calibrated to the aggregate unemployment rate, 4.9 percent over 2014 to 2024 in the Current Population Survey, and is tilted towards low productivity households with the unemployment rates by education of the Bureau of Labor Statistics. The job finding rate is 0.55, from Krueger et al. (2014) (Appendix A, Block 9). A SUD and SMI do not raise separations. Each lowers the chance of finding work, by a divisor calibrated to the share of the homeless with that condition, 21.7 percent for a SUD and 25 percent for SMI (Gutwinski et al., 2021), since a household out of work longer is likelier to lose its home (Table 3, Blocks 2 and 3). Homelessness makes separations 20 percent more frequent and job finding 20 percent slower, from Desmond and Gershenson (2016).

The unemployment rate given a SUD is no longer a target. It is measured in the 2024 NSDUH for adults aged 26 and over, the ages the model covers, at 16.2 percent, and the model gives 14.5 percent without being fitted to it (Table 4, Block 2). One miss remains, and the paper reports it: 21.4 percent of the homeless are employed in the model, against 46 percent in Meyer et al. (2024) (Table 4, Block 1).

All of this is in Section 4.8, in Tables 3, 4 and A1, and in Appendix A, Blocks 9 and 10.

M1(c) The income to drug use relationship is not shown

> *"... the relationship between income and drug use (severe or mild) is key to the calibration of b and the income-dependent penalties in the model ... but its empirical counterpart does not feature very prominently in the paper. It would be helpful to get a sense of this joint distribution in the data."*

Done, and it is now a figure. Figure 2 (Section 5.2, Block 2) plots the share of households that use drugs at each level of income relative to the mean, in the model and in the 2024 NSDUH. As in the data, use is highest at the bottom and lowest at the top: 18.3 percent of households with less than 0.22 of mean income use drugs, against 13.7 percent in the data, and 4.2 percent

of those with more than 1.62 times the mean, against 4.8 percent. Between 0.43 and 1.62 times the mean, use in the model is flatter and lower than in the data, 5.8 to 6.3 percent against 7.2 to 10.2. Both differences lower the average income of users in the model, which is why it falls short of its target: users earn 62 percent of the average in the model, against 74 percent in the data (Table 3). The gradient matters for the experiments because the vouchers are means tested: the lower users sit in the income distribution, the more of them, and of their children, the screen can reach.

On severity, a user is a household whose use amounts to a disorder of any severity, mild included, and a household with a SUD is one whose disorder is moderate or severe. Use that never reaches a disorder is not modelled. Figure 2 is drawn for users in this sense. **[Still open.]** The paper does not show the gradient separately for moderate or severe disorders.

What the gradient disciplines has changed. In the previous draft it was key to b and to penalties that depended on income, as you say. There are no income dependent penalties now: a SUD changes the same hazards by the same factor at every income. The gradient pins one parameter, the mortality multiplier of a SUD $\kappa_{\phi d}$, and b is set to the participation rate, 8.90 percent. The reason is the result of the stylized model (Section 3.3, Figure 1). Drugs are a luxury good bought at a price, so without any consequence of use the average user would earn 3.51 times the population average. The mortality risk of a SUD alone takes this to 0.64 at its calibrated strength, because dying cannot be insured against and what it destroys grows with income. The unemployment risk alone leaves it at 3.51, because a household can save against it; with saving switched off, the unemployment risk alone takes it to 0.75. The gradient in the data is therefore informative about the mortality channel in particular, and that is the parameter it is used for.

This is in Section 3.3 and Figure 1, in the $\kappa_{\phi d}$ item of Section 5.1 (Block 1), and in Section 5.2 (Block 2, Figure 2).

Major concern 2: modelling of the drug addiction process

M2(a) Reducing the disorder mechanically raises the utility of consumption

> *"... reducing addiction lowers θ_d ... and thus increases directly consumption utility from housing and non-housing, thus implying mechanically that policies reducing addictions and policies raising housing consumption are strong welfare complements even absent incentive effects. I am not sure there is an alternative model that would make more sense, but I would like the author to acknowledge and potentially provide more sensitivity to this modelling choice."*

[Conceded.] You are right, and Referee 2 raises the same point (C5). The weight on the drug no longer changes with the disorder. Flow utility, equation (12) in Section 4.2, is the same function of consumption, housing and the drug in every state: θ and b are the same with and without a SUD, with and without SMI, and housed or homeless, and SMI does not enter utility at all (Section 4.7). The stylized model has the same property (equations (2) and (3)), and no preference parameter varies with income either.

Reducing the disorder therefore no longer raises the utility of consumption and housing by itself. A SUD makes a household worse off only through what the model measures: a higher risk of dying, a lower chance of finding work, a higher chance of SMI, the earnings penalty its child inherits, which the parent counts at its weight on the child, and, under Treatment First, the risk

of losing its voucher. The one utility term that depends on the disorder is the cost of a year in treatment, which only a household with a SUD can incur, and only if it chooses treatment.

What remains of a complementarity between housing and recovery is a transition probability, not a taste. Recovery is less likely while homeless, by the multiplier μ_H, which targets the effect of housing on abstinence in the randomised trial of Milby et al. (2005), 1.47 in the data against 1.45 in the model (Table 3, Block 4). The same trial randomised housing conditional on abstinence against housing without the condition, and the model comes close to its ratio without targeting it, 1.40 against 1.26 (Table 4, Block 8).

With no preference parameter that varies with the state, there is no longer a choice of this kind to vary, so the paper does not report a sensitivity analysis on it. On the curvature of utility, the results of the stylized model hold with log utility as well, recalibrated to the same targets (a footnote in Section 3.1).

The preferences are in Section 4.2, equation (12).

M2(b) What patterns does the Stone Geary form actually generate?

> *"I find the use of the Stone-Geary functional form to generate a large mass of non-users, combined with income-dependent penalties ... very clever. However, I am unclear on what are the actual patterns of drug consumption this generates, and where heterogeneity in v (conditional on addiction status d) stems from. It would be very helpful to present model-derived stylized facts on drug use ... by income, employment status, housing status, assets ... and explain how they depend on model parameters (notably b, the Stone-Geary parameter, and $\aleph$, the value of life)."*

The new Section 3 answers the first half of this, and Section 5.2 the second.

How the functional form works. The marginal utility of the drug is proportional to $\theta/(b+v)$ (Section 3.1). With $b > 0$ the marginal utility of the first unit is finite, so a household with little to spend buys none. b therefore mostly decides whether a household uses at all, and θ how much a user uses, that is, the share of its spending that goes on the drug. The risk of a SUD is the same whatever the quantity (equations (4) and (17)), so it sits on the decision to use, not on how much.

Where heterogeneity in v comes from, given d. Among households that use, how much they use moves with what they have to spend, through θ: income, assets and the voucher. Whether they use at all moves with what using risks, and that differs across households with the same d: a household with more to lose from dying is deterred more, as in M1(c). Across values of d, the difference is the risk itself. A household without a SUD that uses risks acquiring one, while one that already has a SUD risks nothing further by using in the benchmark, since mortality, job finding and recovery depend on the disorder and not on the quantity used (equations (17) to (21)).

What the model generates. The paper reports these patterns against the data:

- by income: Figure 2 (M1(c));
- by age: 67 percent of users are 26 to 34 in the model, against 45 percent in the NSDUH for 2023, and their mean age is 34.6 against 39.6, so users in the model are younger than in the data (Table 4, Block 2);
- by housing: 20.2 percent of the homeless have a SUD in the model, against 21.7 percent in the data, a target (Table 3, Block 2);

- by mental health: households with SMI are 2.64 times as likely as others to use, against 2.44 (Table 4, Block 2);
- by employment: 14.5 percent of households with a SUD are unemployed, against 16.2 percent (Table 4, Block 2);
- by parenthood: 5.82 percent of children live with a parent who has a SUD, against 4.6 percent in the 2024 NSDUH (Table 4, Block 6), and 16.35 percent live with a parent who uses drugs (Table 5).

[Still open.] The paper does not report drug use by assets, or by employment status among households without a SUD, and it does not show how these patterns move with ℵ.

b and ℵ. b is set to the participation rate, 8.90 percent; as it falls, the first unit is worth more and more households use (Section 5.1, Block 1). ℵ is no longer a free choice. It targets a value of a statistical life (VSL) of 80 times average labour income (Viscusi and Aldy, 2003), and the model gives 81 (Table 3). As it rises, dying costs more and use falls. Because ℵ enters utility additively and utility is concave, the VSL rises faster than consumption, roughly with its square at the model's risk aversion, as in Hall and Jones (2007), and this is why the mortality cost of a SUD grows with income and puts use at the bottom. The calibration also checks that no household has a negative value of life, so death is never optimal. Figure 1, in the stylized model, shows how who uses changes as the mortality risk of a SUD, whose cost ℵ sets, is scaled from zero to twice its calibrated strength, with b recalibrated at every point so that only who uses changes, not how many.

This is in Sections 3.1 and 3.3 and Figure 1, in Section 5.1 (Block 1), and in Section 5.2 (Block 2, Figure 2 and Table 4).

M2(c) Where do the externalities come from if drug use is rational?

> *"Since drug use is 'rational' and agents are forward-looking, the model should make clearer where externalities arise that generate welfare-improving government interventions. There are potentially many sources of fiscal externalities . . . as well as potentially less obvious non-fiscal ones . . . It would be helpful to make those appear analytically or numerically, and establish how optimal policies change when some of this channels are shut down."*

You are right that if households are rational and forward looking, a case for intervention needs something they do not count. The paper looks for it in two places and finds it in the second, and the title changed because of the answer.

Moral hazard and the budget. Housing First insures a household against losing its home when drug use leads to a SUD, unemployment and homelessness, so part of the cost of using falls on the government's budget. In the model that cost runs through the taxes a household stops paying when a SUD keeps it out of work, the unemployment insurance (UI), food assistance and vouchers it draws, and the treatment the government pays for (equation (26)). These are most of the fiscal externalities you list, and they are the case critics of Housing First usually make. Section 6.2 tests them alone. I remove the child from welfare and from the parent's preferences and make its productivity independent of what the parent did, so that a household's drug use can harm only itself and the budget. Taxes are held, and the budget is balanced by the probability of drawing a voucher, π_V. Treatment First then cuts drug use by 28 percent, and yet a newborn loses 1.17 percent of lifetime consumption with the drug price and π_V held, 1.09 percent when π_V spends the screen's savings on more vouchers, and 0.74 percent when the drug price clears

as well (Table 5). When π_V clears, 58 percent of households gain, but only by 0.05 percent of consumption on average, while 96 percent of those with a SUD lose. The fiscal externality is real, but correcting it is worth less than the option the screen takes from households that choose optimally.

Two items on your list are not in the model. The government's budget has no separate service cost per homeless household: a shelter is free to the household, and equation (26) has no term for it. Since Treatment First lowers homelessness in the long run and along the main transition, such a cost would only add to what the screen saves. And the loss of consortium is gone (m2). A disamenity from the size of the homeless population, which Referee 2 asked about, is in a footnote to Section 2: at the value Loubière et al. (2020) find, it moves no long run gain in Table 5 by more than 0.14 percentage points.

The children. What decides the comparison is the parent's imperfect altruism. The parent values its child's flow utility at $\omega_c(\xi(j) - 1)$ (equation (12)), and what its disorder does to the child's adult value at ω_c, through the charge $D(y, d)$ (equations (24) and (25)). The welfare criterion counts the child at its full weight, $\xi(j) - 1$ (equation (31)), so the share $1 - \omega_c$ is external. ω_c is 0.085, the weight at which removing the parent's concern for its child raises drug use in the model by as much as the 2024 NSDUH shows: unmarried adults living with one other adult and no child are 17.5 percent more likely to have used illicit drugs in the past year than lone parents living with one child, reweighted to the same age, sex, marital status and family income (Section 5.1, Block 5). Three externalities run through that gap:

- *the budget externality*: the child shares its parent's consumption and dwelling but not the drug, so every unit the parent spends on the drug, and every unit of earnings it loses while a SUD keeps it unemployed, is consumption and housing the child goes without;
- *the earnings externality*: a child whose parent had a SUD while it lived at home starts adult life with productivity lower by $\delta_d = 0.24$ (Nilsson, 2017), keeps it for life, and passes a lower draw to its own child (equation (22));
- *the childhood homelessness externality*: a child lives wherever its parent lives. This is the one critics of Treatment First expect it to make worse. In the long run it falls, from 1.03 to 0.92 percent of children in a shelter, and it does not rise along the main transition.

[New result.] Numerically. Section 6.3 splits a newborn's gain exactly into four parts (equation (32) and Table 6). In the main experiment the newborn gains 2.57 percent of lifetime consumption: 1.50 percent from the childhood its parent does not count (the budget and childhood homelessness externalities), 0.46 percent from better productivity at entry that its parent does not count (the earnings externality), 0.43 percent from its own adult life under the policy, and 0.19 percent that its parent does count. Three quarters of the gain is external. The parents gain as well, by 0.69 percent of their own lifetime consumption.

The steps of Table 5 switch the channels on one at a time, which is the exercise you propose:

Newborn's gain, percent of lifetime consumption	Both held	π_V clears	Price clears	Both clear
Without the externalities	−1.17	−1.09	−0.89	−0.74
With the externalities	2.82	6.45	0.98	2.57

"Both held" keeps the drug price and π_V at their Housing First levels; without the externalities the child appears only as an adult, so the gain is its value at 26. The externalities turn a loss into

a gain in every column. Spending the screen's savings on vouchers is worth little without them and a great deal with them, because the savings are larger (π_V rises to 9.24 percent rather than 6.81) and the extra vouchers house children. The drug price works against Treatment First: the screen lowers demand, the price falls, and the cheaper drug draws users back, which takes the gain from 6.45 to 2.57 percent. Single channels are switched off in the text. With the earnings penalty at zero, Treatment First still raises welfare (a footnote in Section 2). With ω_c at 0.69, the upper end of the literature (Nishiyama, 2000), the newborn gains 3.18 percent and 23 percent of it is external (Section 5.1, Block 5).

I keep households rational, as the paper did before. With present bias, a sobriety condition could be worth something to a household as a commitment device even with no externality at all (Section 2), so the assumption works against Treatment First.

[Still open.] The paper does not solve for an optimal programme. It compares the two policies the debate is about and shows which channel decides that comparison.

This is in Sections 4.2 and 4.10 for the parent's weight, Section 6.2 for moral hazard alone, and Section 6.3, equation (32) and Tables 5 and 6 for the decomposition.

M2(d) How is the price elasticity targeted when p_v is normalised to one?

> *"I am somewhat unclear how the elasticity of drug use to its price is targeted, given that the price of drugs p_v is normalized to 1. The author says they vary the price of drugs in the model, but this requires suppressing this normalization and potentially affects the definition of other prices in the model?"*

You are right that a price cannot be fixed and varied at once, and the previous draft did not separate the two. The paper now does.

The price is an equilibrium price. It clears the drug market against a supply curve, $p_v = \nu_1 S_v^{1/\varepsilon_s}$ (equation (27), Section 4.12). The normalisation is confined to the scale ν_1, which is set so that the price is one in the benchmark and then held fixed across experiments. It fixes the units in which the drug is measured and nothing else (Appendix A, Block 13). Each experiment either holds the price at its benchmark level or lets it clear (Table 5), and the transition lets it clear every period.

Moving it touches no other price. Consumption is the unit of account in the budget constraint, equation (16); the rent per unit of housing is held at its benchmark value in every simulation; the interest rate is exogenous; and earnings are productivity times an age profile. No other price in the model is defined relative to the drug. The suppliers are outside the economy, so what they earn along the supply curve enters neither welfare nor the budget (a footnote in Section 4.12).

The demand elasticities are no longer targeted. They are the response of drug use to its price in the benchmark, and they are checks (Section 5.2, Block 5, and Table 4). The participation elasticity is -1.32, against about -0.3 to -0.9 in Saffer and Chaloupka (1999) and Grossman and Chaloupka (1998), and the intensive elasticity among users is -0.63, against about -0.3 to -0.8 in Grossman and Chaloupka (1998) and Olmstead et al. (2015). Participation is more elastic in the model than in the data, and that works against Treatment First: as Treatment First lowers the drug price, more households return to using, and the more elastic participation is, the more of them do. The response to the screen itself, at a given price, is a separate object, matched to the deterrence of the US military's random drug testing (Mehay and Pacula, 1999) through the

detection rate (Section 6).

The supply elasticity. $\varepsilon_s = 1.46$, the estimate of Mace et al. (2020) for cannabis on Washington State administrative data (Section 5.1, Block 6). I know of no direct estimate for an illicit drug. Cannabis is the closest market I could find: it is still a Schedule I substance federally, so its suppliers operate under federal prohibition, much as an illicit supplier does. It is a short run estimate, over a month; longer windows give higher elasticities, and studies of illicit drugs take the long run supply elasticity to be infinite (Becker et al., 2006). The lower the elasticity, the more the price falls when Treatment First lowers demand, and the more users it draws back, so a short run value works against Treatment First. At 1.46 the price falls by 5 percent in the main experiment.

This is in Section 4.12, equation (27); Section 5.1, Block 6; Appendix A, Block 13; and Section 5.2, Block 5, with Table 4.

Minor concerns

m1 The labour supply elasticity should matter more

> *"I am surprised that there is not a clearer role for the labor supply elasticity in welfare calculations, as it should matter quite substantially for the welfare costs of balancing the budget with extra labor taxes."*

You are right that if a policy were financed by a higher labour income tax, the labour supply elasticity would matter for its welfare cost. [**Still open.**] The paper no longer has that margin, and I should say so plainly: labour is supplied inelastically when employed (Section 4.1). I solved earlier versions of this model with endogenous labour supply, and also with endogenous rents, homeownership and house prices. These margins had a negligible effect on the results, so I removed them (a footnote in Section 4.4). The Associate Editor asked in the same round what role labour supply plays in the framework (A3), and for this question the answer is very little.

It matters less than you expected because of how the experiments are financed. No experiment raises taxes. The tax parameters are held at their benchmark values in every experiment, and the budget is balanced by the probability of drawing a voucher, π_V (Section 4.11; Appendix A, Block 11). And Treatment First does not need financing. Users who lose their vouchers leave the programme, and tax revenue rises, mainly because more children start adult life with higher productivity, so π_V rises, from 6.75 to 7.80 percent in the main experiment. The deadweight cost of a higher labour tax does not arise in any experiment in the paper.

The labour market still matters, through employment rather than hours: a SUD lowers the chance of finding work, and homelessness lowers it and raises separations (Section 4.8). Those are what move tax revenue.

This is in Section 4.1, the footnote in Section 4.4, Section 4.11 and Appendix A, Block 11.

m2 Present sensitivity as graphs rather than tables

> *"Some of the results of the counter-factual experiments might be more helpfully presented in the form of graphs comparing outcomes and sensitivity to parameters, rather than tables, for example, for the sensitivity to the loss of consortium spillover, or to the value-of-life parameter ℵ."*

The loss of consortium is gone. Referee 1 argued (3(a) and 3(b)) that an adjustment made after the model is solved is inconsistent with it, and that the conservative value of a parameter without a source is zero. I agree, and the paper no longer has one. What a household's drug use costs others is now inside the model, in the parent's imperfect weight on its child (M2(c)).

The transition, where outcomes move over time, is a figure: Figure 3 has eight panels that set the path against both steady states year by year. The long run (Table 5), the decomposition (Table 6) and the alternative transitions (Table 7) are tables, because each compares a handful of discrete experiments.

[Still open.] The paper does not plot the results against ℵ. ℵ is no longer a free choice: it is calibrated to a VSL of 80 times average labour income (Viscusi and Aldy, 2003). The one figure that varies a parameter continuously is Figure 1, in the stylized model, which scales the mortality and unemployment risks of a SUD from zero to twice their calibrated strength and shows who uses at each point. The sensitivity checks of the quantitative model each move one thing to one alternative value, so the paper reports them in the text, mostly in footnotes:

- the parent's weight on its child at 0.69 instead of 0.085, with b recalibrated to drug use: the newborn gains 3.18 percent instead of 2.57 (Section 5.1, Block 5);
- the detection rate at 10 percent instead of 20: the newborn gains 1.65 percent instead of 2.57 (Section 6);
- the earnings penalty at zero: Treatment First still raises welfare (Section 2);
- a homelessness disamenity at the value of Loubière et al. (2020): no long run gain moves by more than 0.14 percentage points (Section 2);
- households with a SUD unable to keep their voucher by entering treatment: the newborn gains at least as much (Section 6.2);
- each alternative transition at two detection rates, 20 and 100 percent (Table 7).

These are in Figure 1, Figure 3, Tables 5 to 7, and the footnotes listed above.

m3 Vary the voucher receipt probability continuously

> *"It would be helpful to provide a more continuous experiment that varies the probability of voucher receipt rather than go straight for full rationing to fully universal provision. This would help establish whether there is a Laffer curve of voucher generosity, or whether the corner solution of universal provision is indeed optimal, and what are the key parameters governing this."*

[Still open.] The paper no longer compares rationed with universal vouchers, so the sweep you suggest is not in it. The reason is the question. The policy now on the table, the July 2025 executive order and the HUD funding notice that followed it (Section 1), is about whether the assistance that exists should require treatment and sobriety, not about how much of it there is. That assistance is rationed: 26.6 percent of income eligible households hold a voucher. Every experiment therefore keeps the programme rationed and asks whether to condition it (Section 6).

The voucher probability is not fixed, though. Unless an experiment holds it, π_V adjusts to keep the government's net revenue at its benchmark level, so it moves as far as the reform's savings allow (Section 4.11): from 6.75 to 7.80 percent in the main experiment, to 9.24 percent with the drug price held, and between 6.55 and 6.90 percent in the first decade of the transition (Section 6.4). Comparing the columns of Table 5 where π_V is held and where it clears shows what that extra generosity is worth. With the drug price held, spending the screen's savings on vouchers raises a

newborn's gain from 2.82 to 6.45 percent; without the externalities, the same step moves it only from a loss of 1.17 percent to a loss of 1.09. Whether there is a Laffer curve, or universal provision is optimal, is a question about the scale of the programme, and the paper does not answer it.

This is in Section 4.11, Section 6.2 and Table 5, and Section 6.4.